Two-point spectral calibration and the remaining insertion freedom

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Abstract

We combine the fixed-mass recoil limit of the direct C_0 factor with the observer density and isolate an exact finite-mass spectral factor. Large-mass matching leaves a family of positive, symmetric insertion normalizations unfixed. Fixed mean mass, fixed initial mass and a thresholded trace integral give different spectra. We derive their relations and compute a declared normalization model, including its threshold-induced t^{-2} tail. Neither that model nor its finite-mass corrections are identified as a unique gravitational correlator.

1 The established QFT calibration

We work in dS_3, with radius one and principal-series scalar parameter $\nu > 0$, so that the field mass squared is $1 + \nu^2$. The no-defect limit of CLP (3.38)–(3.40) agrees with CSTY (1.16)–(1.17) [1, 2]:

$$k_\nu(\omega) = \frac{1}{8\pi[\cosh(\pi\omega) + \cosh(\pi\nu)]}, \tag{1}$$

$$g_\nu(t) = \int_{\mathbb{R}} \mathrm{d}\omega\, e^{-i\omega t} k_\nu(\omega) = \frac{\sin(\nu t)}{4\pi \sinh(\pi\nu) \sinh t}, \qquad g_\nu(0) = \frac{\nu}{4\pi \sinh(\pi\nu)}. \tag{2}$$

There is no additional $1/(2\pi)$ in this forward transform. This is the antipodal, two-sided function at Euclidean separation $\theta = \pi + it$, not a same-side Wightman function. Its spectral evenness is therefore expected.

Our order of limits remains fixed-mass $\kappa \to \infty$, followed by any large-mass limit of the resulting function. The fixed-cone-angle heavy-heavy-light prescription for the insertion normalization $\mathcal{N}$ in CLP (3.37) does not supply its full fixed-mass recoil limit. We do not assume a gravitational Hilbert space or physical four-point sewing rule.

2 Assembling the known factors

After removing a common mass-independent factor, the recoil limit of the observer weight CLP (3.10) is

$$z(m) = m^2 e^{-2\pi m}, \qquad m > 0. \tag{3}$$

The previous fixed-mass derivation gave the paired direct-C_0 factor, up to a common constant,

$$D(\boldsymbol{\mu}) = \frac{\pi e^{\pi S} \prod_a [\sinh(\pi\mu_a)/\mu_a]}{\cosh(\pi S/2) \prod_a \cosh(\pi(S - 2\mu_a)/2)}, \qquad S = \sum_a \mu_a. \tag{4}$$

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Set $m_{1,2} = M \pm \omega/2$, and abbreviate

$$A = \cosh(2\pi M), \qquad B = \cosh(\pi\omega), \qquad C = \cosh(\pi\nu).$$

Direct hyperbolic product identities yield

$$\frac{z(m_1)z(m_2)D(m_1,m_2,\nu)}{z(M)} = 16\pi^2 \frac{e^{\pi\nu}\sinh(\pi\nu)}{\nu}\, k_\nu(\omega) R_M(\omega), \tag{5}$$

$$R_M(\omega) = \left(1 - \frac{\omega^2}{4M^2}\right)\frac{A-B}{A+C}. \tag{6}$$

In this convention the constant

$$n_0(\nu) = \frac{\nu e^{-\pi\nu}}{16\pi^2 \sinh(\pi\nu)} \tag{7}$$

aligns the bounded-ω, large-M limit with k_ν. It is not an absolute determination of the original $\mathcal{N}$: common constants in z, D have already been removed.

Write the remaining unknown insertion function as $n(M,\omega)$. A candidate energy kernel then has the form

$$\mathcal{Q}_n(m_1,m_2) = z(M) k_\nu(\omega) R_M(\omega) n(M,\omega). \tag{8}$$

Positive, even n gives a positive, label-symmetric real kernel. These properties do not establish an operator or gravitational inner-product realization. Below $n = 1$ is a declared diagnostic completion, not an identified physical prescription.

For a fixed lower mass $m_* > 0$, let $D_M = 2(M - m_*)$. For $n = 1$, $0 \le R_M \le 1$ on the integration interval, and

$$\sup_{t\in\mathbb{R}} \left| \int_{-D_M}^{D_M} \mathrm{d}\omega\, e^{-i\omega t} k_\nu R_M - g_\nu(t) \right| \le \frac{g_\nu(0)(1+\nu^2)}{12M^2} + \frac{D_M}{4\pi(A+C)} + \frac{e^{-\pi D_M}}{2\pi^2}. \tag{9}$$

Indeed $1 - R_M \le \omega^2/(4M^2) + (B+C)/(A+C)$, $\int \omega^2 k_\nu = g_\nu(0)(1+\nu^2)/3$, and $k_\nu \le e^{-\pi|\omega|}/(4\pi)$. This is an absolute bound, including zeros of g_ν. At fixed finite M the hard endpoints instead give a late-time term $2k_\nu(D_M)R_M(D_M)\sin(D_M t)/t$. Thus (9) is not a uniform relative late-time claim.

3 Normalization freedom survives calibration

For any $\lambda \ge 0$, consider

$$n_\lambda(M,\omega) = 1 + \lambda \frac{\omega^2}{4M^2}. \tag{10}$$

It is positive and even on the real domain, agrees at $\omega = 0$ for every M, and tends to one at fixed ω. Nevertheless,

$$R_M n_\lambda = 1 + (\lambda - 1)\frac{\omega^2}{4M^2} + O_{\nu,\omega}(M^{-4} + e^{-2\pi M}). \tag{11}$$

The leading finite-mass correction can therefore be changed or removed while retaining all these calibration conditions. The $n = 1$ correction cannot yet be advertised as a unique physical prediction. Determining n needs additional physical normalization or sewing information.

4 Fixed initial mass and the thermal factor

Adopt the energy convention $E = -m$ of CSTY, but explicitly choose $m_1 = m$ as the initial mass and $m_2 = m - \omega$ as the final mass. Then $\omega = E_f - E_i$ and the phase for $W(t)W(0)$ is $e^{-i\omega t}$. The conditional candidate spectrum is

$$w_{m,n}(\omega) = \frac{\mathcal{Q}_n(m, m-\omega)}{z(m)}, \qquad \omega \le m - m_*. \tag{12}$$

Label symmetry implies the exact reversed-transition identity

$$z(m)w_{m,n}(\omega) = z(m-\omega)w_{m-\omega,n}(-\omega). \tag{13}$$

It relates different initial states and is not the KMS condition for one finite-m state. For $n = 1$, the two polynomial factors combine to $1 - \omega/m$:

$$w_{m,1}(\omega) = e^{\pi\omega}k_\nu(\omega)\left(1 - \frac{\omega}{m}\right)\frac{\cosh\pi(2m-\omega) - \cosh\pi\omega}{\cosh\pi(2m-\omega) + \cosh\pi\nu}. \tag{14}$$

If $n \to 1$ at fixed transfer and large mass, then

$$w_\infty(\omega) = e^{\pi\omega}k_\nu(\omega), \qquad \frac{w_\infty(-\omega)}{w_\infty(\omega)} = e^{-2\pi\omega}. \tag{15}$$

This recovers the standard inverse temperature 2π. The exact density ratio in (13) is $(1 - \omega/m)^2 e^{2\pi\omega}$; its local entropy slope $2\pi - 2/m$ must not be mistaken for exact same-state thermality.

The time-domain statement is best made inside the analytic strip:

$$g_\nu(t + i(\pi - \epsilon)) = \int_{\mathbb{R}} \mathrm{d}\omega\, e^{-i\omega t}e^{(\pi-\epsilon)\omega}k_\nu(\omega), \qquad 0 < \epsilon < 2\pi. \tag{16}$$

The $\epsilon \downarrow 0$ boundary is the same-side Wightman distribution, with short-distance singularity $-i/[4\pi(t - i0)]$. Its positive-frequency spectrum approaches a constant. Consequently the two-sided L^1 tail argument cannot be applied to an unregulated Wightman integral.

5 A thresholded trace model

Now explicitly set $n = 1$ and integrate both masses above $m_0 > 0$. This defines a spectral trace model; no gravitational operator trace is established by this definition. With $\beta = 2\pi$,

$$Z_0 = \int_{m_0}^\infty z(m)\ \mathrm{d}m = e^{-\beta m_0}\left(\frac{m_0^2}{\beta} + \frac{2m_0}{\beta^2} + \frac{2}{\beta^3}\right), \tag{17}$$

$$s_{m_0}(\omega) = \frac{k_\nu(\omega)}{Z_0}\int_{m_0+|\omega|/2}^\infty z(M)R_M(\omega)\ \mathrm{d}M. \tag{18}$$

The spectrum is positive and even, as expected for a tracial protocol. This is distinct from (15). The threshold mechanism and its Cauchy convolution already appear in CSTY (1.12)–(1.14); here we retain the m^2 weight and the full factor R_M.

Let

$$c_0 = \frac{m_0 + 1/\beta}{m_0^2 + 2m_0/\beta + 2/\beta^2}. \tag{19}$$

Dropping only the exponentially small part of R_M gives an exact elementary mass integral,

$$s^{(0)}_{m_0}(\omega) = k_\nu(\omega)e^{-\pi|\omega|}(1 + c_0|\omega|), \tag{20}$$

$$0 \le 1 - \frac{s_{m_0}(\omega)}{s^{(0)}_{m_0}(\omega)} \le 2(1 + \cosh\pi\nu)e^{-2\pi m_0}. \tag{21}$$

The relative bound is uniform on the whole real frequency axis.

For an exact convergent correction assume $m_0 > \nu/2$, and set

$$b_j = (j+2)\beta, \quad L = m_0 + |\omega|/2, \quad q_j = (-1)^j \frac{\sinh((j+1)\pi\nu)}{\sinh(\pi\nu)}.$$

Expanding the quadratic denominator in $e^{-\beta M}$ gives

$$s_{m_0} = s^{(0)}_{m_0} - \frac{1}{4\pi Z_0} \sum_{j=0}^{\infty} q_j e^{-b_j L} \left[\frac{m_0^2 + m_0|\omega|}{b_j} + \frac{2m_0 + |\omega|}{b_j^2} + \frac{2}{b_j^3} \right]. \tag{22}$$

For $m_0 \le \nu/2$ one must instead use (18) or another valid representation.

The moving lower limit produces a zero-frequency cusp:

$$s'_{m_0}(0+) = -\frac{k_\nu(0) z(m_0) R_{m_0}(0)}{2Z_0}.$$

The one-sided derivatives through fourth order are integrable. Integration by parts therefore yields

$$T_{m_0}(t) := \int_{\mathbb{R}} e^{-i\omega t} s_{m_0}(\omega)\, \mathrm{d}\omega = \frac{k_\nu(0) z(m_0) R_{m_0}(0)}{Z_0 t^2} + O(t^{-4}). \tag{23}$$

It is a threshold effect in this model, not a finite-G chaos exponent. A general smooth even insertion n changes the cusp coefficient by $n(m_0, 0)$, provided the required integrability holds.

6 Checks and the four-point target

The final record contains 71 checks. Algebraic factor assembly, the reversed-transition relation, the interior-strip Fourier transform and detailed balance are checked directly. The normalization family (11) is also evaluated: the scaled first correction tends to $-1, 0, +1$ for $\lambda = 0, 1, 2$.

For $\nu = 0.6$, $m_0 = 1, 3$, we compute the actual Fourier integral of $s^{(0)}$, and transform each exponential-polynomial term of (22) analytically into a rational function of time. The curves are not generated from (23). At $t = 128$, the ratios to the leading tail are 1.00028507 and 0.99994557, respectively.

Frequency quadrature uses orders 28/40, precisions 45/65 digits, step 0.1, and cutoff 12, extended to 14 at the largest time. The mass integral is independently compared with the 20/24-term series. The omitted series contributes, uniformly in time, at most

$$\frac{e^{-2\beta m_0} \rho^N}{2\pi Z_0 (1 - e^{-2\pi\nu})(1 - \rho)} \left[\frac{2m_0^2}{b_N^2} + \frac{8m_0}{b_N^3} + \frac{8}{b_N^4} \right], \qquad \rho = e^{\pi\nu - \beta m_0} < 1. \tag{24}$$

The baseline frequency tail has a separate elementary exponential bound. Their combined bound relative to the computed signal is at most 2.05×10^{-28} in this sample. The largest quadrature refinement is 2.61×10^{-43}; this much smaller number is not a total error bar. Roundoff and quadrature errors are not interval-certified. The initial 59-check version is retained; 12 additional checks cover truncation bounds and cutoff extension.

For two independent free probe fields, the fixed-mean, large-observer-mass, $G = 0$ antipodal disconnected target is

$$F_{\mathrm{disc}} = g_\nu(0) g_{\nu'}(0) = \frac{\nu\nu'}{16\pi^2 \sinh(\pi\nu) \sinh(\pi\nu')}. \tag{25}$$

This is a Wick-contraction calibration, not a graph value at zero energy transfer. Identical fields have additional Wick pairings, and a shared finite-mass constraint need not factorize.

The next step is to obtain an actual insertion/sewing rule, including its spectral measure, and test this disconnected target. Two-point matching is necessary but does not determine crossing dynamics. An identity-leg limit must also be treated correctly: $\nu = 0$ still has field mass squared one; an identity dimension requires a non-principal continuation toward $\nu = \pm i$ and possibly a distributional limit. Finite G and the complete de Sitter dual remain open.

Reproducible records. Chinese derivation, code and raw diagnostics are saved at:

notes/stage4-recoil-spectral.md
code/recoil_spectral.py
results/stage4/recoil-spectral/diagnostics.json

The diagnostic record includes normalization variants, precision comparisons and input hashes.

References

[1] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1, (3.10), (3.14), (3.35)–(3.40).

[2] Y. Chen, D. Stanford, H. Tang and Z. Yang, *Negative shocks versus static patch holography*, arXiv:2607.14042v1, (1.4)–(1.17) and (2.1)–(2.3).